



Software Requirements Specification for Aktour ViaBalkan

Project Team:

Furkan Duman, Dogukan Yurtturk, Eren Çapar, Can Aysen

Date: March 31, 2026

Version: 1.0

Course: Software Engineering CS308

Revision History

Name	Date	Reason For Changes	Version
Project Team	15.03.2026	Initial document structure created and draft started.	0.1
Project Team	24.03.2026	Developed system features, external interfaces, and non-functional requirements.	0.5
Project Team	31.03.2026	Initial draft created combining project proposal and SRS template structure.	1.0

Contents

1	Introduction	2
1.1	Purpose	2
1.2	Document Conventions	2
1.3	Intended Audience and Reading Suggestions	2
1.4	Project Scope	2
1.5	References	2
2	Overall Description	2
2.1	Product Perspective	2
2.2	Product Features	2
2.3	User Classes and Characteristics	3
2.4	Operating Environment	3
2.5	Design and Implementation Constraints	3
2.6	User Documentation	3
2.7	Assumptions and Dependencies	3
3	System Features	3
3.1	Secure Login and Role-Based Access Control	3
3.2	Admin Dashboard and Data Visualization	4
3.3	Tour Creation Wizard	4
3.4	Expense Management and Receipt Upload	4
3.5	Guide, Vehicle, and Route Management	4
4	External Interface Requirements	4
4.1	User Interfaces	4
4.2	Software Interfaces	4
4.3	Communications Interfaces	5
5	Other Nonfunctional Requirements	5
5.1	Performance Requirements	5
5.2	Security Requirements	5
5.3	Software Quality Attributes	5
A	Glossary	5

1 Introduction

1.1 Purpose

This document specifies the software requirements for Aktour ViaBalkan, a desktop application designed to manage tourism operations in the Balkan region. It is intended to provide a clear and structured description of the system functionalities and constraints for developers and stakeholders.

1.2 Document Conventions

This document follows a standard IEEE-based SRS structure. Priority levels (High, Medium, Low) are assigned to system features to guide the iterative development cycle and sprint planning.

1.3 Intended Audience and Reading Suggestions

This document is intended for:

- **Developers & Architects:** To guide the frontend UI/UX architecture, backend API development, and system integration.
- **Instructors & Teaching Assistants:** To evaluate the project scope, system design, and adherence to requirements for the CS308 course.
- **Stakeholders:** To understand the proposed functionalities for tour managers, administrators, and guides.

1.4 Project Scope

Aktour ViaBalkan addresses the problem of tour companies relying on manual calculations and generic management tools that lack region-specific data for the Balkans. The system will automate expense tracking, simplify tour management, and provide real-time dashboards to analyze tour profitability. It allows administrators to create tours, calculate tour costs (fuel, tolls, guide fees), track vehicles, and manage tour-related expenses.

1.5 References

- Course: Software Engineering CS308 Project Proposal: Aktour ViaBalkan.

2 Overall Description

2.1 Product Perspective

Aktour ViaBalkan is a new, self-contained client-server desktop application. It is built with JavaFX for the user interface and Spring Boot for the backend services. The system connects to a MySQL database hosted on an Ubuntu server.

2.2 Product Features

- Secure login system with role-based access.
- Admin dashboard with charts and financial data visualization.
- Tour creation wizard with dynamic cost calculation.
- Expense management with receipt upload.
- Guide, vehicle, and route management.

2.3 User Classes and Characteristics

- **System Administrators:** Manage tours, vehicles, and guides, monitor expenses, and analyze financial charts.
- **Tour Guides:** Update tour information, manage profiles, and upload receipts or expense documents.
- **Developers:** Responsible for server maintenance, database reliability, and API security.

2.4 Operating Environment

- **Frontend:** Desktop operating systems running JavaFX.
- **Backend Server:** Ubuntu Server.
- **Database:** MySQL.

2.5 Design and Implementation Constraints

- The frontend must be built utilizing JavaFX and Scene Builder.
- The backend must expose a REST API utilizing Spring Boot.
- Version control must be maintained via GitHub utilizing Conventional Commits.

2.6 User Documentation

- System design documentation and testing report.
- Final project report and presentation.

2.7 Assumptions and Dependencies

- **Assumptions:** The team must maintain precise Balkan-specific data for tolls and route distances to ensure calculation accuracy.
- **Dependencies:** The system depends on reliable communication and integration between the JavaFX frontend and the Spring Boot backend via the API.

3 System Features

3.1 Secure Login and Role-Based Access Control

- **3.1.1 Description and Priority:** High priority. A secure authentication system to enforce role-based access.
- **3.1.2 Functional Requirements:**
 - **REQ-1:** The system shall provide a secure login mechanism using unique credentials.
 - **REQ-2:** The system shall enforce role-based access control, ensuring users only access authorized data.

3.2 Admin Dashboard and Data Visualization

- **3.2.1 Description and Priority:** High priority. A dashboard providing administrators with financial visualizations.
- **3.2.2 Functional Requirements:**
 - **REQ-3:** The system shall display financial charts summarizing tour profitability on the administrator dashboard.

3.3 Tour Creation Wizard

- **3.3.1 Description and Priority:** High priority. A step-by-step wizard for creating tours and generating cost estimates.
- **3.3.2 Functional Requirements:**
 - **REQ-4:** The system shall calculate dynamic tour costs (fuel, tolls, guide fees) based on region-specific Balkan data.

3.4 Expense Management and Receipt Upload

- **3.4.1 Description and Priority:** Medium priority. A module for logging expenses and uploading supporting documentation.
- **3.4.2 Functional Requirements:**
 - **REQ-5:** The system shall allow tour guides to upload receipts and expense documents.

3.5 Guide, Vehicle, and Route Management

- **3.5.1 Description and Priority:** Medium priority. Interfaces to manage logistical profiles and tracking.
- **3.5.2 Functional Requirements:**
 - **REQ-6:** The system shall allow administrators to manage vehicle and guide profiles.

4 External Interface Requirements

4.1 User Interfaces

- The application shall provide a modern and intuitive graphical user interface built with JavaFX and Scene Builder.
- The UI must minimize delay and facilitate easy navigation for administrative and guiding tasks.

4.2 Software Interfaces

- **Database:** The system shall interface with a MySQL database to store users, tours, vehicles, and expenses.
- **Backend:** The JavaFX client shall communicate with the Spring Boot backend via RESTful APIs.

4.3 Communications Interfaces

- All data transfer between the desktop client and the Ubuntu server will be handled over standard HTTP/HTTPS protocols utilized by the REST API.

5 Other Nonfunctional Requirements

5.1 Performance Requirements

- The system shall ensure fast API responses with minimal UI delay.

5.2 Security Requirements

- The system shall utilize role-based access control (RBAC) to ensure proper data security and system integrity.
- API endpoints must be secured to protect operational tourism data.

5.3 Software Quality Attributes

- **Usability:** The interface must be highly intuitive, mitigating the learning curve required to master the new platform.
- **Scalability:** The underlying architecture must be designed to allow for future expansion and the integration of new modules.

A Glossary

- **API / REST API:** Application Programming Interface used for communication between the JavaFX frontend and the Spring Boot backend.
- **JavaFX:** The software platform utilized for creating the desktop user interface.
- **RBAC (Role-Based Access Control):** Restricting system access based on user roles (e.g., Administrator vs. Tour Guide).
- **Spring Boot:** The Java-based framework utilized for building the backend services.
- **Sprint:** A set period of time during which specific work (e.g., implementing the Expense Module) must be completed and made ready for review.